

Bank Customer Churn Analysis Using Microsoft Excel

Prepared by

Pragati Gaikwad
Aspiring Data Analyst

1. Executive Summary

Customer churn is a critical business issue in the banking industry because retaining existing customers is generally more cost-effective than acquiring new ones. Understanding the factors that influence customer attrition enables organizations to improve customer satisfaction, increase retention, and reduce revenue loss.

This project analyzes a Bank Customer Churn dataset using Microsoft Excel. Through data preprocessing, Pivot Tables, Pivot Charts, calculated metrics, and an interactive dashboard, the analysis identifies key churn drivers and provides actionable business recommendations.

2. Project Objective

The primary objective of this project is to analyze customer churn patterns and identify the characteristics of customers who are most likely to leave the bank.

The project aims to:

- Calculate the overall customer churn rate.
 - Identify high-risk customer segments.
 - Analyze demographic, financial, and behavioral factors influencing churn.
 - Build an interactive Excel dashboard.
 - Provide business recommendations for improving customer retention.
-

3. Business Problem

Customer attrition directly impacts profitability, customer lifetime value, and long-term business growth.

The bank seeks answers to the following business questions:

- What is the overall customer churn rate?
- Which customer segments are most likely to churn?

- Does age influence customer churn?
 - Does customer engagement reduce churn?
 - Does product ownership affect retention?
 - Which customers should receive proactive retention efforts?
-

4. Dataset Description

The dataset consists of **10,000 customer records** containing demographic, financial, and behavioral information.

Dataset Features

- CustomerId
- CreditScore
- Geography
- Gender
- Age
- Tenure
- Balance
- NumOfProducts
- HasCrCard
- IsActiveMember
- EstimatedSalary
- Exited

Target Variable:

- Exited = 1 → Customer Churned
 - Exited = 0 → Customer Retained
-

5. Tools & Techniques Used

The project was completed entirely using Microsoft Excel.

Techniques used include:

- Pivot Tables
 - Pivot Charts
 - GETPIVOTDATA()
 - IF() Functions
 - Conditional Formatting
 - Dashboard Design
 - KPI Cards
 - Interactive Slicers
-

6. Methodology

Step 1 – Data Collection

The Bank Customer Churn dataset was obtained from Maven Analytics.

Step 2 – Data Preprocessing

New analytical columns were created to simplify business analysis.

Examples include:

- Age Groups
 - Salary Categories
 - Balance Categories
 - Credit Score Categories
 - Tenure Categories
-

Step 3 – Data Cleaning

The dataset was reviewed for consistency.

The following checks were performed:

- Missing values
 - Data formatting
 - Category validation
 - Numerical consistency
-

Step 4 – Exploratory Data Analysis

Customer demographics and financial attributes were explored using Pivot Tables.

Multiple business questions were developed before beginning the analysis.

Step 5 – Data Analysis

Fifteen business analyses were performed using Pivot Tables and Pivot Charts.

The analyses included:

1. Overall Customer Churn
 2. Gender Analysis
 3. Geography Analysis
 4. Age Group Analysis
 5. Credit Card Ownership
 6. Customer Engagement
 7. Balance Analysis
 8. Product Analysis
 9. Credit Score Analysis
 10. Salary Analysis
 11. High-Value Customer Analysis
 12. Geography & Gender Analysis
 13. Age & Product Analysis
 14. Customer Tenure Analysis
 15. Customer Churn Persona Development
-

Step 6 – Dashboard Development

An interactive Excel dashboard was developed featuring:

- KPI Cards
- Interactive Charts
- Business Insights
- Business Recommendations
- Interactive Slicers

The dashboard allows users to explore customer churn across multiple dimensions including age, geography, credit score, product ownership, salary, and engagement status.

7. Key Findings

Overall Customer Churn

- Total Customers: **10,000**
 - Churned Customers: **2,037**
 - Overall Churn Rate: **20.40%**
-

Age Analysis

Senior customers experienced the highest churn rate among all age groups, indicating that customer age plays an important role in customer retention.

Customer Engagement Analysis

Inactive members exhibited a significantly higher churn rate (26.9%) than active members (14.3%).

This suggests that customer engagement is one of the strongest indicators of churn.

Credit Card Analysis

Customers with and without credit cards displayed very similar churn rates.

Credit card ownership appears to have little influence on customer retention.

Customer Tenure Analysis

Customer churn remained relatively stable across tenure categories.

Tenure alone is not a reliable predictor of customer churn.

High-Risk Customer Segment

The highest-risk customer segment identified during the analysis was:

Inactive Members (26.9% Churn Rate)

This customer group should be prioritized for retention initiatives.

8. Dashboard Overview

The final dashboard summarizes key business metrics using:

- Total Customers
- Churned Customers
- Overall Churn Rate
- Highest Risk Segment
- Interactive Charts
- Key Insights
- Business Recommendations
- Dynamic Slicers

The dashboard enables business users to quickly identify churn trends and monitor customer behavior.

(Insert Dashboard Screenshot Here)

9. Business Recommendations

Based on the analysis, the following recommendations are proposed:

Increase Customer Engagement

Develop personalized engagement campaigns targeting inactive customers.

Improve Senior Customer Retention

Design loyalty programs and customized financial products specifically for senior customers.

Monitor High-Risk Customers

Develop churn monitoring dashboards to identify customers showing early signs of disengagement.

Strengthen Customer Relationships

Implement proactive communication strategies before customers decide to leave.

Support Data-Driven Decision Making

Use regular customer churn reporting to evaluate the effectiveness of retention strategies.

10. Project Outcome

This project demonstrates how Microsoft Excel can be used to perform end-to-end business analysis.

Key accomplishments include:

- Data preprocessing
- Exploratory Data Analysis
- Business problem solving
- Pivot Table reporting

- Interactive dashboard creation
- Business storytelling
- Data-driven recommendations

The project highlights practical Excel skills commonly required for entry-level Data Analyst positions.

11. Future Enhancements

Future improvements to this project may include:

- Building an interactive Power BI dashboard.
 - Creating predictive churn models using Python.
 - Applying Machine Learning algorithms to predict customer churn.
 - Developing SQL-based reporting pipelines.
 - Automating dashboard refresh using Power Query.
-

12. Conclusion

The analysis successfully identified important factors associated with customer churn.

Among all variables analyzed, customer engagement and age emerged as the strongest indicators of customer attrition. These findings can help banking organizations develop targeted retention strategies, reduce customer churn, and improve long-term profitability.

This project demonstrates how business problems can be solved effectively using Microsoft Excel and analytical thinking.